Yuchen Liu

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AI systems engineer building RL training infrastructure, search/data pipelines, and research-grade open-source tools.

EDUCATION

M.S. Computer Science	University of Pennsylvania, Philadelphia	GPA: 3.80/4.00	May 2025
B.S. Applied Economics	Pennsylvania State University, State College	GPA: 3.86/4.00	May 2023

WORK EXPERIENCE

Software Engineer, Search Infrastructure | DoorDash — Aug 2025 - Present | Seattle, WA

- Built Notus, a next-generation search ingestion platform using Iceberg, Kafka, and Spark to decouple document ingestion from serving, enable configuration-driven indexing, and unblock schema evolution for hybrid search.
- Built Vela, Argo's hybrid retrieval platform, enabling Lucene-native hybrid search and federated retrieval with vector backends such as Milvus behind a single API.
- Migrated 30+ production stacks to modular Jsonnet, eliminating roughly 56,000 lines of duplicate configuration and reducing config verification time by 90%.
- Stabilized production indexers with 500GB EBS storage, pre-deployment validators, and Kyverno guardrails after RCA for $21K revenue-risk incidents, preventing capacity-related outages.

Software Engineering Intern | Penn Medicine TissueLab — Oct 2024 - May 2025 | Philadelphia, PA

- Developed a Python/FastAPI AI microservice platform to orchestrate PyTorch models for medical image analysis, enabling natural-language-driven segmentation and classification workflows.
- Implemented an event-driven DAG workflow engine for async task execution, processing 10,000+ pathology images daily and streaming 12M+ imaging events through WebSockets with sub-10ms latency.
- Integrated generative and discriminative deep learning models into a unified service layer with asynchronous REST APIs for cross-department usage.

Software Engineering Intern | Information Technology of CAS — Apr 2024 - Aug 2024 | Remote

- Designed Kafka and Redis-backed event streaming services that reduced p95 API latency from 2s to 200ms while processing millions of IoT events.
- Contributed to distributed Spring Boot monitoring services, delivered 35+ REST APIs, and maintained 99.7% uptime in Unix-based production environments.

OPEN SOURCE

nanoRL | **github.com/RiddleHe/nanochat**

- Led an open-source RL training framework inside nanochat for clean objective definitions, rollout experiments, RL algorithm research, reproducible ablations, and vLLM inference serving.

nanochat | **github.com/RiddleHe/nanochat**

- Built a hackable pretraining and chat stack supporting architecture definition and FLOP-controlled model ablations.

llm-interp | **github.com/RiddleHe/llm-interp**

- Authored reproducible interpretability scripts for model circuit research, including attention-sink analysis and reproductions of LLM decode indeterminism findings.

RESEARCH

Do Value Vectors in Deep Layers Need Context from the Residual Stream? — EMNLP 2026, under review

Co-first author. Challenged the standard attention mechanism of computing value vectors from the residual stream, finding that deep layers benefit from context-free value vectors that preserve original token information.

Proposed Bank of Values, a learned value-vector table for the last third of layers that eliminates the V cache and improves validation loss and average scores across 21 benchmarks on 135M and 780M models. arXiv:2606.02780

Vision Language Models Cannot Plan, but Can They Formalize? — ECCV 2026, under review

Studied whether VLMs can formalize visual planning tasks into solver-executable PDDL problem files. arXiv:2509.21576

A co-evolving agentic AI system for medical imaging analysis — Nature, under review

Built agentic medical-imaging workflows for tool selection, planning, knowledge updates, and human-in-the-loop co-evolution. arXiv:2509.20279

SKILLS

Languages: Python, Kotlin, Java, C++, JavaScript; ML/Infra: PyTorch, vLLM, Spark, Kafka, Iceberg, Delta Lake, FastAPI, Redis, Docker, Kubernetes, AWS, Spring Boot